

EECS 841

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Convolutional and Recurrent Neural Networks: Q&A

Convolutional Neural Networks (CNN)

- **Q1: What is the primary purpose of Convolutional Neural Networks (CNNs), and why are they well-suited for image learning tasks?**

A1: CNNs are primarily used for image learning due to their ability to identify patterns, translation invariance, and the ability to subsample images, reducing the number of parameters needed for processing.

- **Q2: Explain the concept of “hierarchical representation” in the context of CNNs.**

A2: Hierarchical representation refers to how CNNs learn features at different levels of abstraction. Low-level features (edges, lines) are assembled into mid-level parts, which are further combined into high-level objects and scenes.

- **Q3: Describe the three main types of layers in a CNN and their functions.**

A3:

- **Convolutional Layer:** Applies filters to input data to create feature maps, detecting patterns.
- **Pooling Layer:** Downsamples the feature maps, reducing dimensionality and preserving detected features (max pooling) or averaging values (average pooling).
- **Fully-Connected Layer:** Connects every neuron in one layer to every neuron in the next, used for final classification.

- **Q4: What are the benefits of using convolution instead of fully connected layers in image processing?**

A4: Convolutional layers require fewer parameters because they connect to only a local region of the input and use shared weights. This reduces computational complexity and the risk of overfitting.

- **Q5: Explain the concepts of padding and stride in the context of convolutional layers.**

A5:

- **Padding:** Adding extra pixels around the boundary of the input image to control the size of the output feature map.
- **Stride:** The number of rows and columns traversed per slide of the filter, affecting the size of the output feature map.

- **Q6: Describe the difference between max pooling and average pooling.**

A6: Max pooling selects the maximum value from each pooling region, preserving detected features. Average pooling averages the values in each region and is used to downsample the feature map.

- **Q7: How does backpropagation work in CNNs, specifically in convolutional layers?**

A7: Backpropagation in convolutional layers involves computing the gradients of the loss function with respect to the filter weights and input data. The chain rule is used to propagate the error backward through the network, updating the weights to minimize the loss.

Recurrent Neural Networks (RNN)

- **Q1: What are Recurrent Neural Networks (RNNs) and what type of data are they designed to process?**

A1: RNNs are a type of neural network designed to process sequential data by maintaining a hidden state that captures information about previous inputs in the sequence.

- **Q2: What is the purpose of Long Short-Term Memory (LSTM) networks, and how do they differ from traditional RNNs?**

A2: LSTMs are designed to address the vanishing gradient problem in traditional RNNs, allowing them to capture long-range dependencies in sequential data through the use of memory cells and gates.